

Spatial synchrony stacks

Bounded real-PRISM implementation, validation, diagnostics, and pre-CONUS decision gate

Outcome: the repository now has an exact moving-center stack verb, rich reductions, and a separate landscape-similarity diagnostic. The 20 by 20 observed PRISM test computed 400 center landscapes from 80,200 canonical pairs while returning all 160,000 center-to-focal values.

Decision

GO for a second bounded, finite-radius benchmark after adding process-level or native parallelism and chunked Zarr checkpoints. NO-GO for a dense CONUS center-by-focal stack: its quadratic pair count and storage are scientifically unnecessary and operationally prohibitive.

Scientific definition preserved

Cold synchrony uses tmin, each series' own median, dates where both values are at or below their medians, and Spearman correlation. Warm synchrony uses tmax, each series' own median, dates where both values are above their medians, and Spearman correlation. Delta S is cold minus warm. Paired missing-value filtering, average ranks for ties, and min_t=10 are inherited from the validated center-reference kernel.

Input	Observed experiment
File	artifacts/center-pixel-synchrony-walkthrough/intermediate/prism_boulder_2023-11-01_2024-01-30.nc
Provider	PRISM Climate Group
Window	2023-11-01T00:00:00.000000000 through 2024-01-30T00:00:00.000000000 (91 labels)
Grid	20 by 20 (400 moving centers and focal pixels)
Synthetic fallback	No
Fingerprint	sha256:4b4a3c30fbd5ccb2a4f8b5cefc2ca2b127e7bf31d646b27091b354b447df9acb

1. Architecture and API

`v.local_synchrony_stack()` returns the literal scientific object: a Dataset with center, y, and x dimensions. The implementation canonicalizes every pair as $(\min(i,j), \max(i,j))$, computes it once, and gathers it into both endpoint stacks. Self-pairs are retained, so the storyboard's 20 by 20 center block yields exactly 400 layers per focal pixel.

Component	Purpose	Interpretation
local_synchrony_stack	Unreduced center landscapes and focal stacks	Primary science object $S(c,p)$
reduce_synchrony_stack	Distribution, coverage, distance, direction summaries	Maps derived from full stacks
synchrony_landscape_similarity	Adjacent-panel Spearman $Q(A,B)$	Regional-break diagnostic, not pair $S(A,B)$
stack_edges	Canonical undirected edge table	Storage/network utility, not primary object

Symmetry and equivalence

Symmetry is exact under argument reversal because paired filtering, per-series quantiles, joint selection, and Spearman correlation are symmetric. Unit tests cover cold, warm, Delta S, missing values, ties, and `min_t`. The real center landscape was pixelwise equal to the established verb at $1e-7$: True.

Field	Maximum absolute error
cold	0
warm	0
delta	0

Previously audited focal pair

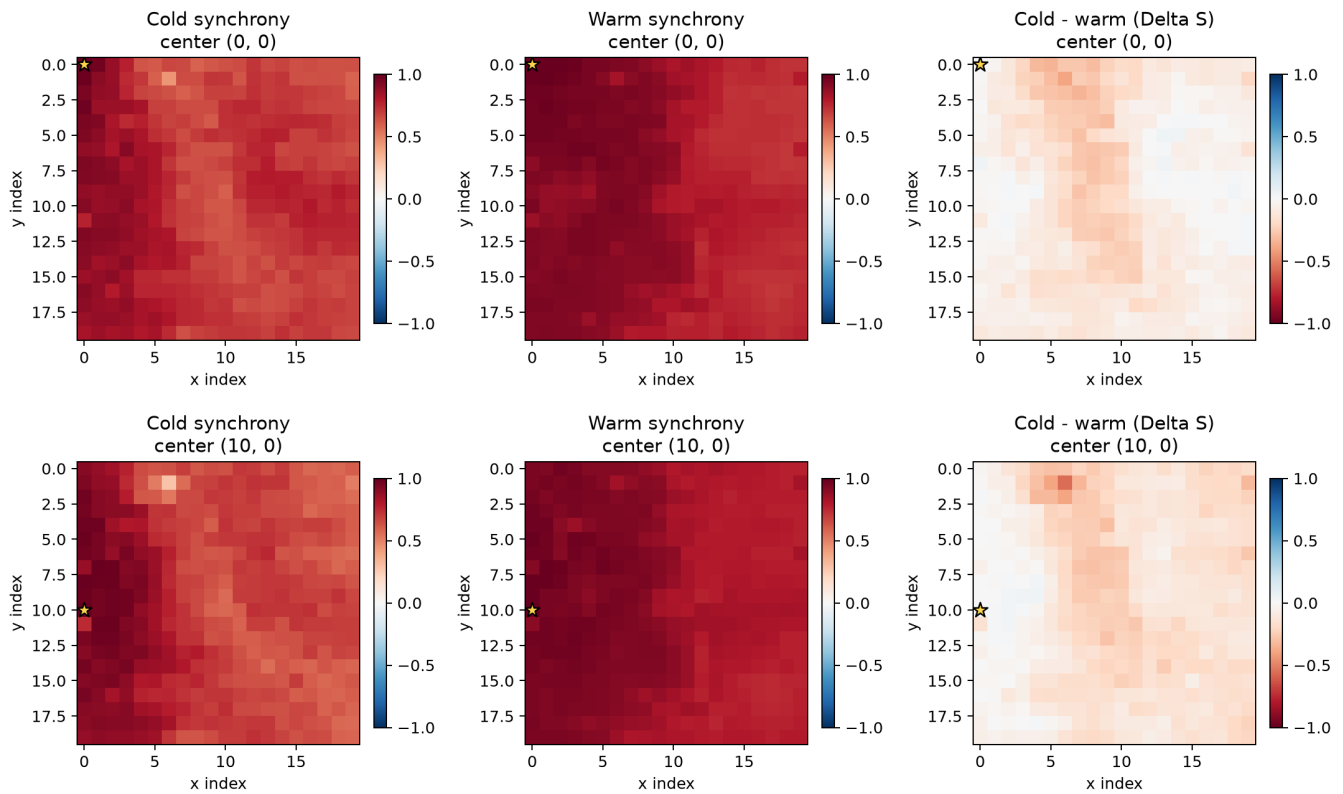
Cold	Warm	Delta S	Distance (km)
0.588180	0.857692	-0.269512	44.838

Expected reference values are approximately 0.588, 0.858, and -0.270; the stack reproduces them.

2. Moving-center landscapes

Each row below is one center landscape. The star marks the center. Blue and red are kept consistent with the signed metric: for Delta S, blue indicates positive cold-minus-warm and red indicates negative cold-minus-warm.

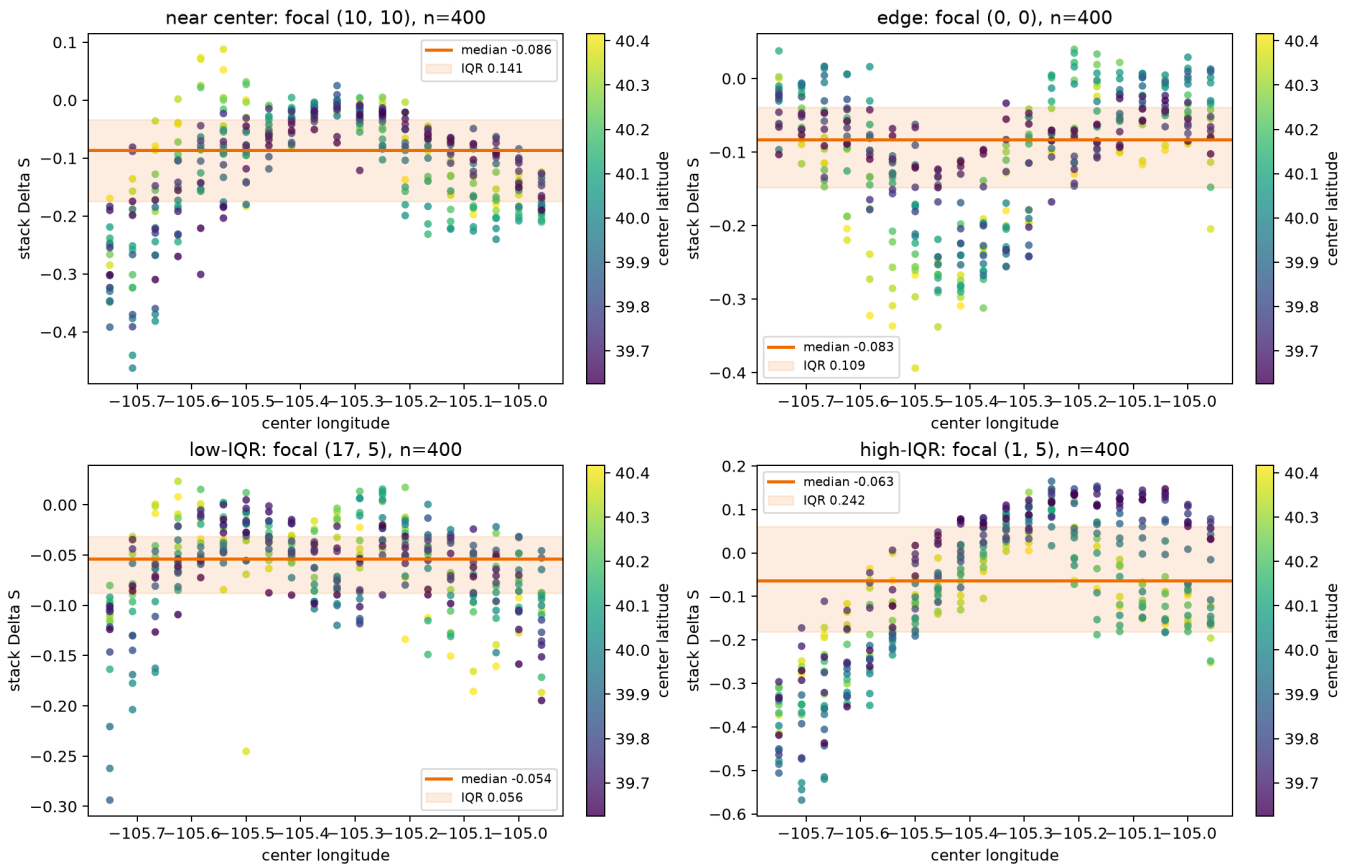
Two contrasting moving-center synchrony landscapes



3. Focal stacks

A focal stack is the vertical slice through all 400 center landscapes at one geographic pixel. The scatter retains center longitude and latitude; the median and IQR are summaries, not replacements for the distribution.

Four focal-pixel stacks retain center-location structure

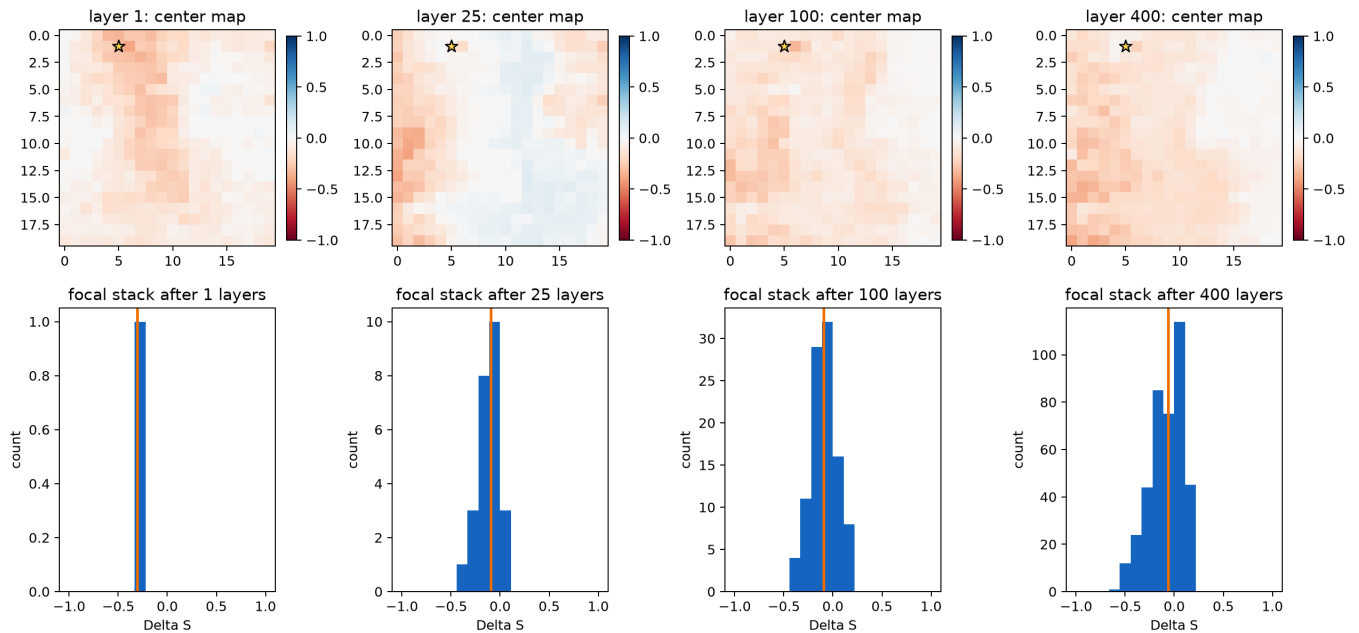


Selected focal roles: near center (10,10); edge (0,0); low-IQR (17,5); high-IQR (1,5). Labels describe selection criteria within this bounded block and are not climate-region classifications.

4. From landscapes to stacks

The upper row shows successive center landscapes; the lower row accumulates the value at one fixed focal pixel. This is the concrete relationship between the moving maps and the eventual focal distribution.

Center maps build the stack at focal (1, 5)

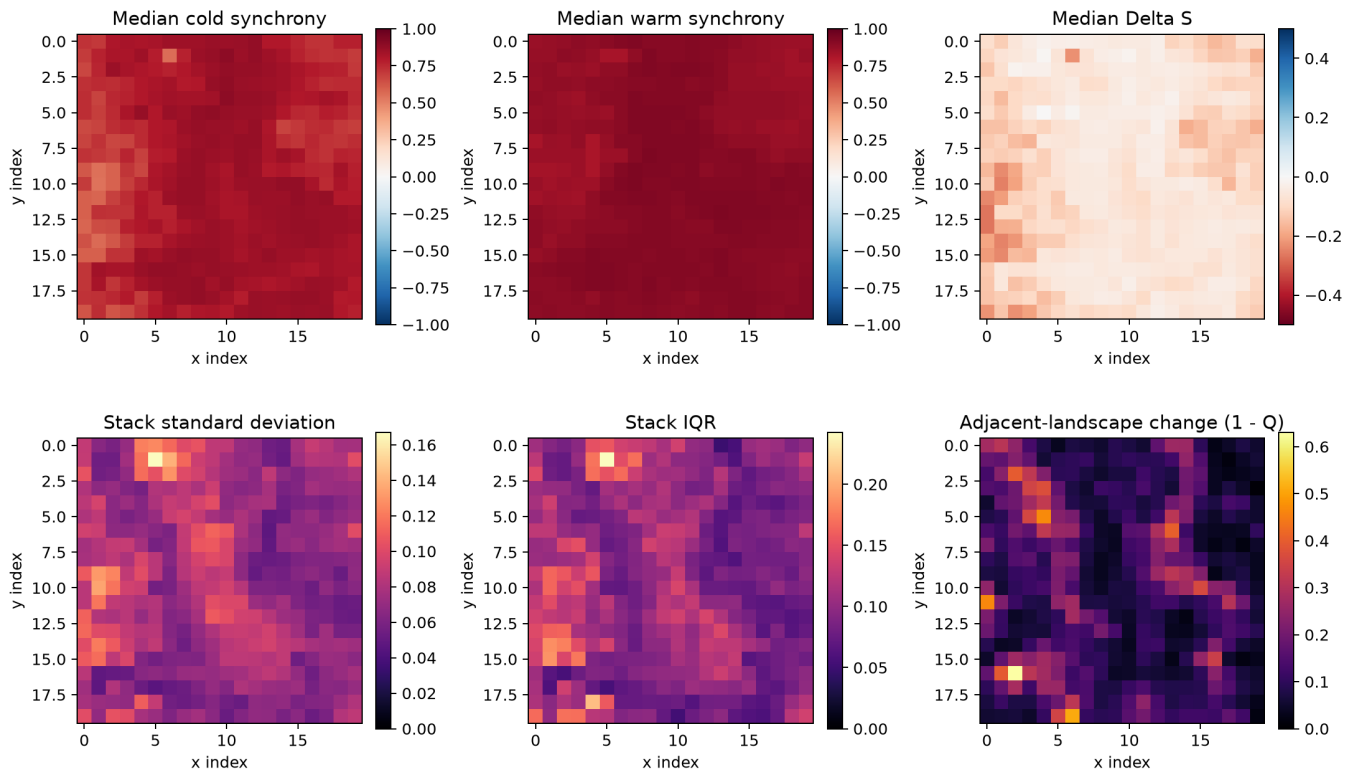


An accompanying GIF repeats this buildup across the block. It is evidence of the data construction, not an inference that the visual groupings are objectively discrete regions.

5. Reductions and regional-break diagnostics

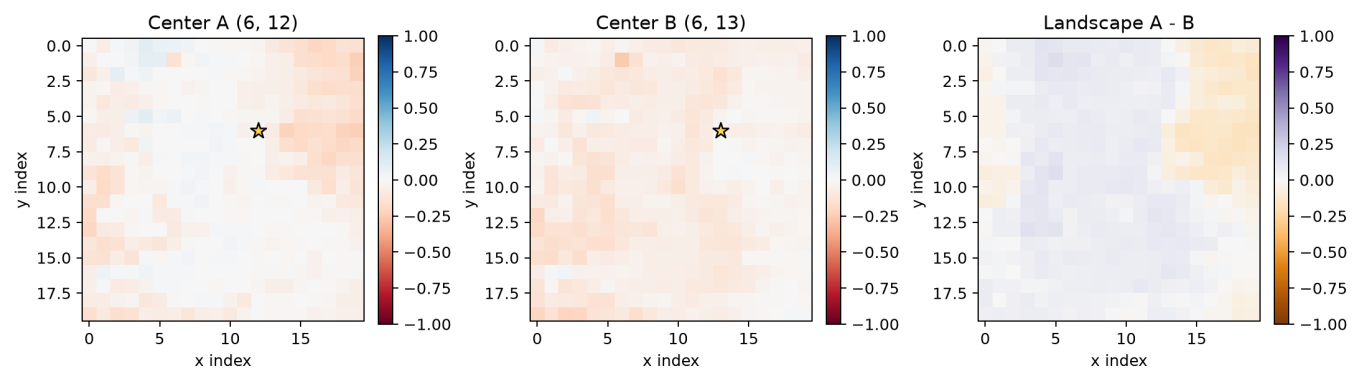
Median cold, median warm, and median Delta S provide compact products. Standard deviation and IQR measure stack spread. The final panel maps 1 - Q averaged over each center's adjacent panel comparisons. Q compares full landscapes; it is not S between the two center pixels.

Reduction maps and a regional-break diagnostic



Most contrasting adjacent landscapes

Strongest adjacent landscape break: panel similarity Q=0.088



Center A	Center B	Q	1-Q	RMSE	Overlap
132	133	0.088	0.912	0.080	400

Boundary caveat: high stack spread or low adjacent-panel similarity can flag a transition worth investigating. It does not by itself prove a true climate region, establish a causal boundary, or validate a clustering model.

6. Benchmark ladder

Timings are single-process wall-clock measurements on this machine. The kernel uses coarse NumPy pair batches and creates zero per-pair Dask tasks. tracemalloc is a Python allocation estimate and may undercount library-level native memory.

Case	Grid	Pairs	Seconds	Pairs/s	Peak MB	Output MB
single nonself pair	1x2	3	0.010	312	0.1	0.0
5 by 5 tile	5x5	325	0.296	1,097	0.2	0.0
20 by 20 block	20x20	80,200	71.024	1,129	23.2	5.0

CONUS resource estimates

Assumption: 910,000 cells at about 4 km. One 91-label tmin+tmax float32 window is about 0.66 GB before masks, coordinates, and overhead.

Support	Approx pairs	1-core hours	Ideal 32-worker hours	Edge GB
50 km radius	223,347,603	54.9	1.7	5.4
100 km radius	893,390,411	219.8	6.9	21.4
250 km radius	5,583,690,068	1,373.6	42.9	134.0
Dense all-pairs	414,050,455,000	101,855	not meaningful	9,937

These are arithmetic planning estimates, not cluster benchmarks. They omit I/O contention, checkpoint metadata, failed-task retries, scheduler overhead, and repeated rolling windows. They do show why exact dense national stacks must not be attempted.

7. Production architecture and restartability

Concern	Phase-one behavior	Production requirement
Spatial work	One bounded 20 by 20 block	Center tiles with finite-radius halos
Pair work	Canonical pairs in coarse batches	Parallel batches; never one task per pair
Output	NetCDF checkpoint plus JSON manifest	Chunked Zarr by window and center tile
Resume	Exact fingerprint and complete-status check	Batch manifest; reject mixed parameters
Failures	Atomic rename after complete write	Retry missing batches; preserve completed compatible batches
Metadata	Indices, coordinates, distance, bearing, bands, counts	Retain unchanged in sparse outputs

Recommended next gate

Benchmark exact 50 km and 100 km neighborhoods on a 100 by 100 observed tile using process-level parallelism and Zarr checkpoints. Compare numerical output with the current kernel at 1e-7, measure real read/write bytes and scheduler tasks, and only then choose a CONUS support radius. If those costs are still excessive, evaluate a documented approximation against exact held-out tiles before any national run.

Limitations

This experiment covers one observed Colorado winter block and one inclusive 90-day coordinate-label interval containing 91 daily labels. It does not establish seasonal stability, national representativeness, climate-region truth, causal mechanisms, or a validated entropy/multimodality metric. Distance-weighted and directional summaries are available, but their scientific interpretation requires a declared analysis question.